

Netflix Catalog Intelligence

Dataset-backed project report · generated October 01, 2026

Executive summary

- International Movies is the largest genre group with 2,752 tagged titles.
- Movie is the most common format (69.7% of the catalog).
- TV-MA is the most frequent audience rating (3,205 titles).
- United States appears most often in country credits (3,240 titles).
- International Movies grew fastest across the observed catalog-add periods (+68600.0%).
- Catalog additions changed -20.3% from 2020 to 2021.
- The median content release year is 2017; this is distinct from when titles were added to Netflix.

The strongest rating classifier on the stratified holdout was XGBoost (accuracy 0.538, weighted F1 0.501). Clustering selected K=2 by silhouette score. Forecast evaluation used a chronological holdout of 12 months.

1. Dataset overview

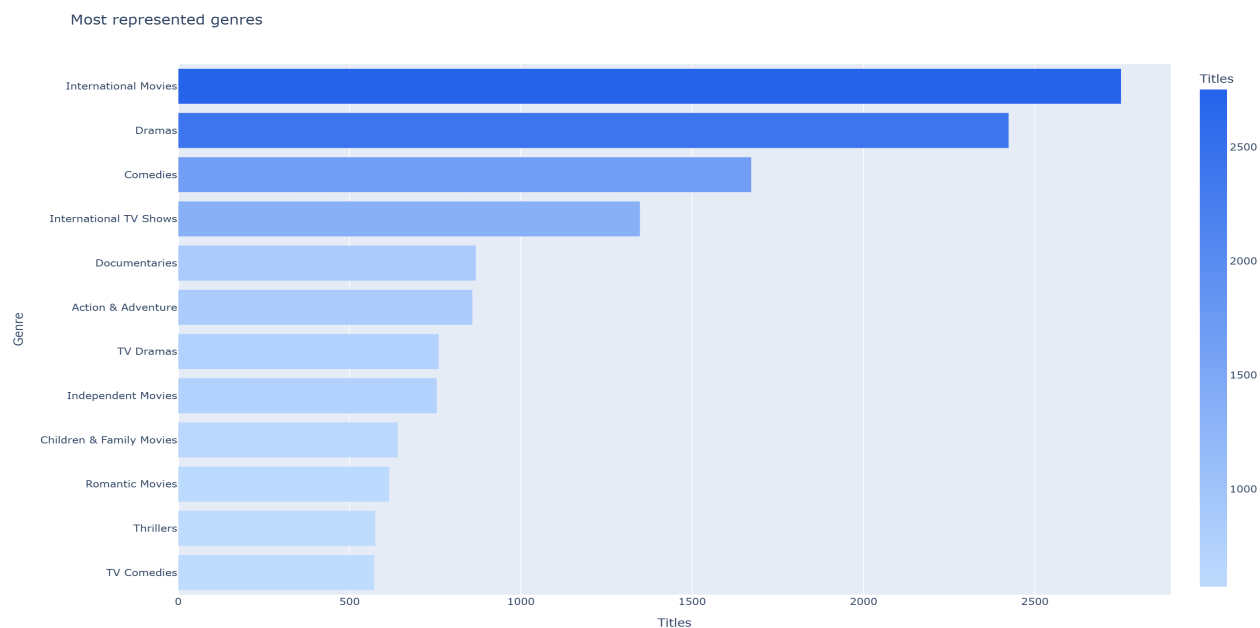
All statistics in this report are computed from the supplied Netflix CSV after shared cleaning and duplicate removal.

Metric	Value
Titles	8,787
Unique titles	8,786
Movie titles	6,124
TV shows	2,663
Unique ratings	14
First release year	1,925
Latest release year	2,021
Rows with catalog-add dates	8,787
Duplicate records removed	3

Missing-value profile

Column	Missing values	Missing %
description	8,787.00	100.00
cast	8,787.00	100.00
title	0.00	0.00
type	0.00	0.00
show_id	0.00	0.00
country	0.00	0.00
director	0.00	0.00
date_added	0.00	0.00
release_year	0.00	0.00
duration	0.00	0.00
rating	0.00	0.00
listed_in	0.00	0.00
director_count	0.00	0.00
cast_count	0.00	0.00

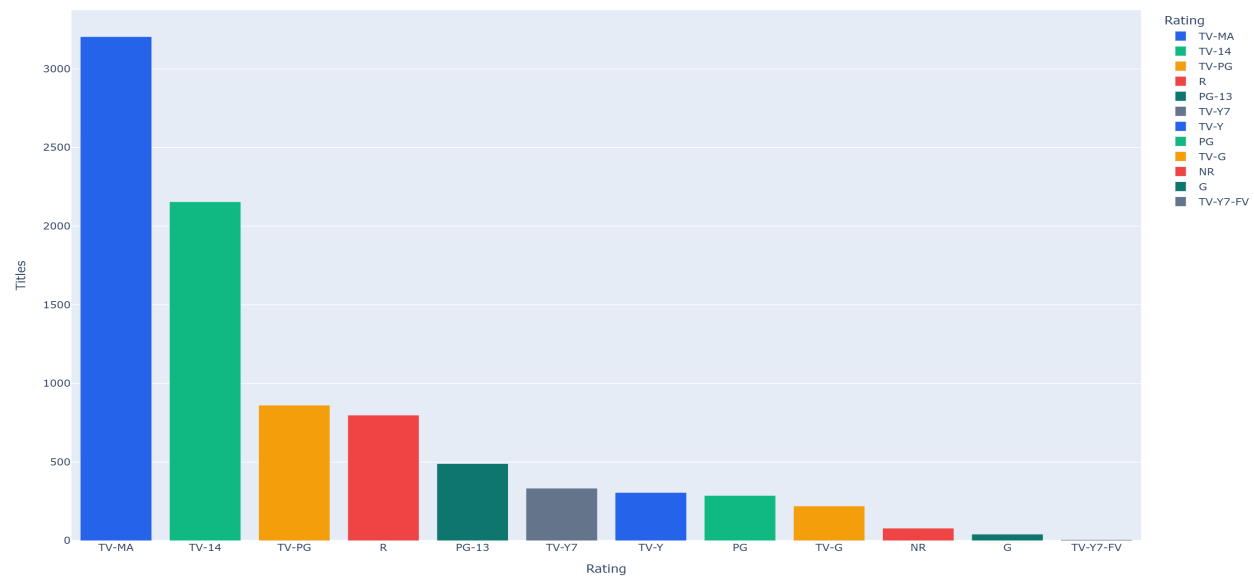
2. Catalog composition



Most credited countries

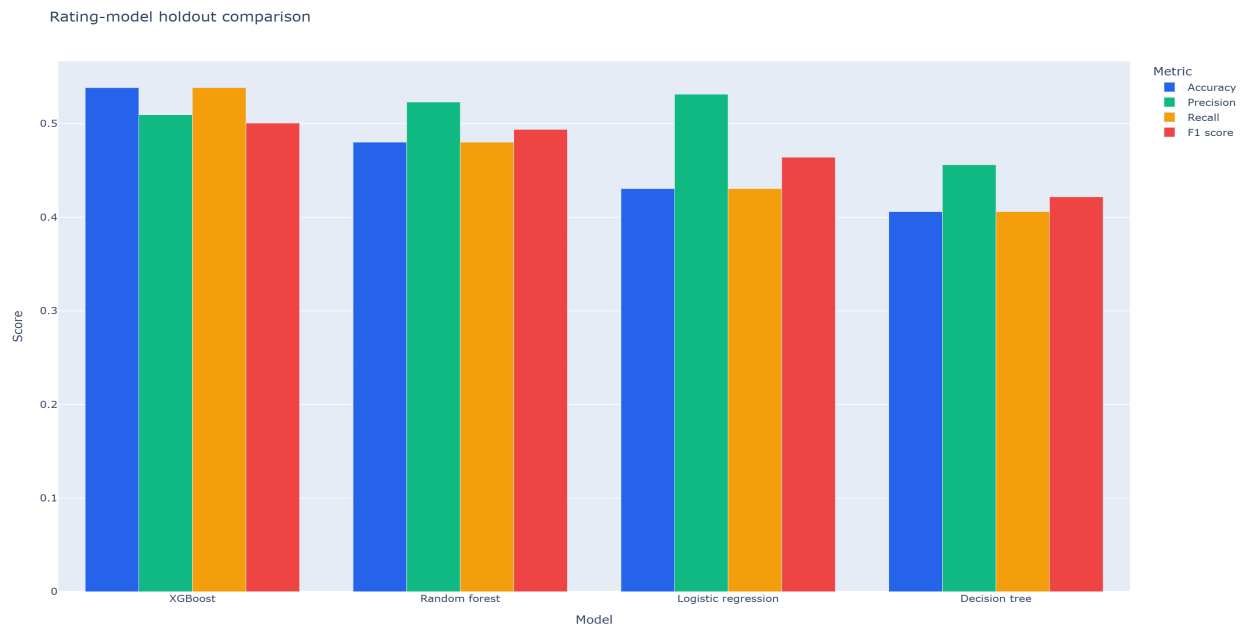
Country	Titles
United States	3,240
India	1,056
United Kingdom	638
Pakistan	420
Not Given	287
Canada	271
Japan	259
South Korea	214
France	213
Spain	182
Mexico	138
Egypt	123

Rating category distribution



3. Rating classification

The target is the dataset's existing content-rating category. This is not a viewership, satisfaction, or popularity label. Evaluation used 6,590 training rows and 2,197 stratified holdout rows.



Model	Accuracy	Precision	Recall	F1 score	Fit seconds
XGBoost	0.538	0.509	0.538	0.501	2.137
Random forest	0.480	0.523	0.480	0.494	1.064
Logistic regression	0.431	0.531	0.431	0.464	0.397
Decision tree	0.406	0.456	0.406	0.422	0.163

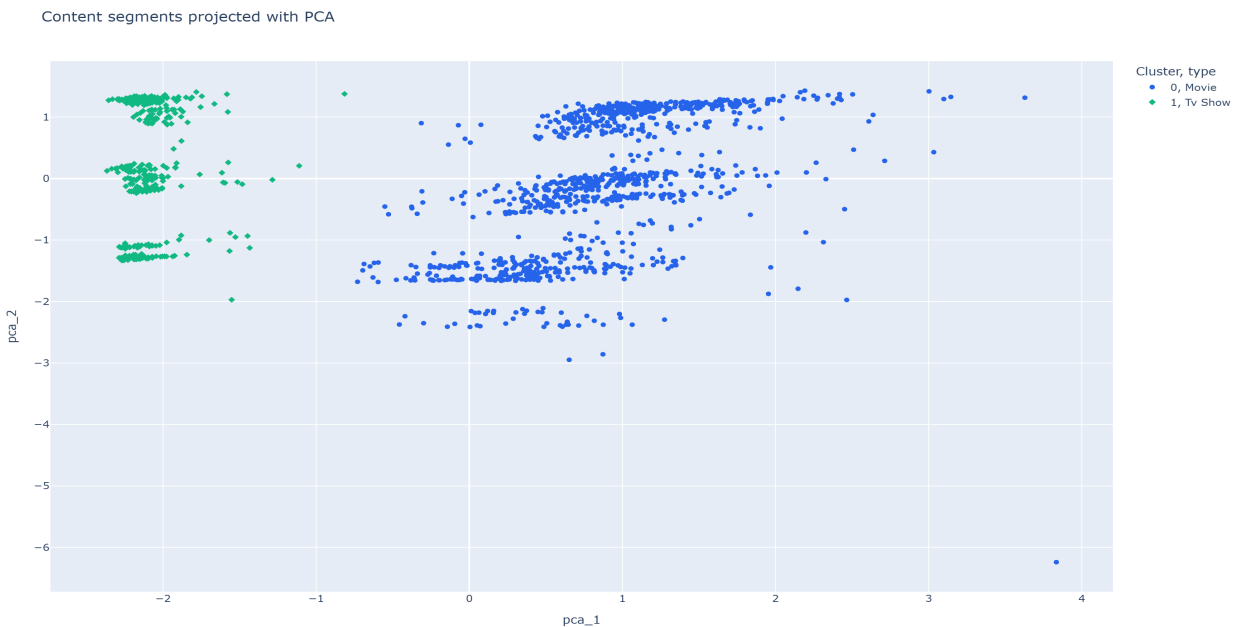
Top encoded feature-importance signals

Feature	Importance
categorical__duration_unit_min	0.0216
categorical__listed_in_Kids' TV	0.0216
categorical__country_primary_Egypt	0.0196
categorical__listed_in_Stand-Up Comedy	0.0172
categorical__country_primary_India	0.0159
categorical__listed_in_Action & Adventure	0.0144
categorical__listed_in_Kids' TV, TV Comedies	0.0126
categorical__listed_in_Children & Family Movies	0.0126
categorical__listed_in_Children & Family Movies, Comedies	0.0124

Feature	Importance
categorical__listed_in_Anime Series, Kids' TV	0.0122
categorical__listed_in_Crime TV Shows, International TV Shows, Spanish Language TV Shows	0.0120
categorical__country_primary_United States	0.0110

4. Content segmentation

KMeans selected 2 clusters by silhouette score. The two-component PCA projection explains 45.0% of the encoded feature variance. Cluster labels are descriptive and should not be treated as official content categories.



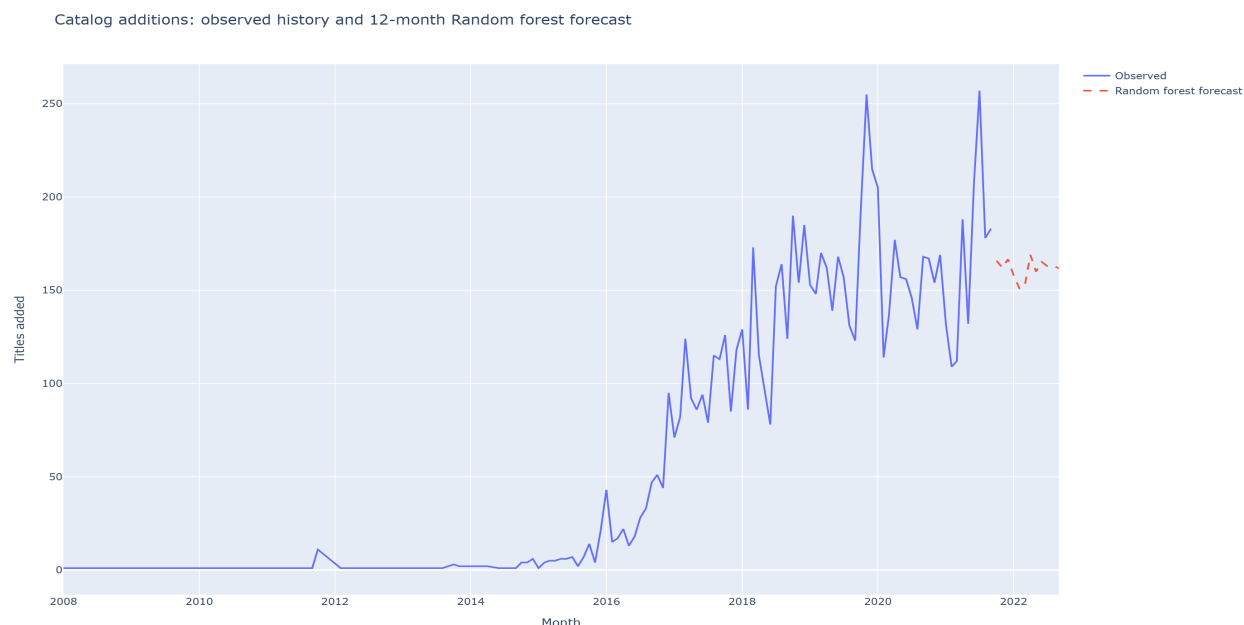
Cluster	Titles	Dominant format	Format mix	Top genres	Average release year
0.0	1,064.0	Movie	Movie (1064)	International Movies, Drama, Comedies	2012.0
1.0	436.0	Tv Show	Tv Show (436)	International TV Shows, TV Dramas, TV Comedies	2006.8

Cluster-count diagnostics

Clusters	Inertia	Silhouette
2.000	7,952.058	0.291
3.000	6,890.047	0.240
4.000	6,094.642	0.251
5.000	5,380.291	0.247
6.000	4,936.109	0.251
7.000	4,570.163	0.228
8.000	4,306.643	0.193

5. Catalog-add forecasting

The monthly series contains 108 observed calendar months. Forecasts model titles added to the catalog, not original production volume or official company plans. The table uses a chronological holdout and reports RMSE, MAE, and R^2 .



Model	RMSE	MAE	R2
Random forest	40.880	31.764	-0.028
ARIMA (1,1,1)	41.490	33.053	-0.059
Prophet	62.244	54.167	-1.383

Forecast insights

- ARIMA (1,1,1) projects approximately 2,213 additions over the next 12 months, a +11.3% change versus the last 12 observed months.
- Prophet projects approximately 2,506 additions over the next 12 months, a +26.1% change versus the last 12 observed months.
- Random forest projects approximately 1,937 additions over the next 12 months, a -2.5% change versus the last 12 observed months.
- Observed additions in the latest 12 months changed -3.1% from the preceding year.

Interpretation and limitations

Content ratings are not success outcomes. Forecasts extrapolate a short historical catalog-add series and should be read with their holdout errors; negative R^2 means the model did not outperform the holdout mean baseline. Genre growth is descriptive, not causal. For business decisions, validate against updated licensed data and monitor model drift.